

2 (a) $3.5 T$ B1 [1]

(b) (i) distance = average speed $\times$ time (however expressed)
= 14 m C1
A1 [2]

(ii) distance = $5.6 \times (T - 5)$ (or $3.5T - 14$) A1 [1]

(c) $3.5T = 14 + 5.6(T - 5)$ C1
 $T = 6.7 \text{ s}$ A1 [2]

(d) (i) acceleration = $(5.6 / 5 =) 1.12 \text{ m s}^{-2}$ C1
force = ma C1
= 75 N A1 [3]

(ii) power = (force $\times$ speed =) $\{75 + 23\} \times 4.5$ C1
= 440 W A1 [2]
(allow 1/2 for 234 W, 0/2 for 338 W or 104 W)

3 (a) (i) potential energy: stored energy available to do work B1 [1]